Diabetes Risk Segmentation & Decision Support System

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Background

You are part of a data science team at a health-tech startup called BC Analytics. BC Analytics partners with clinics, insurers, and wellness programs to improve patient outcomes and reduce long-term healthcare costs. One of their key challenges is the early identification and management of diabetes risk. Currently, healthcare providers often:

- React too late to warning signs
- Lack clear insights into lifestyle drivers of risk
- Do not have tools to support data-driven decisions during consultations

Your mission

You have been tasked with building a data-driven decision support system that will:

1. Segment patients into diabetes risk categories
2. Identify key lifestyle factors driving risk
3. Group patients into meaningful lifestyle-based segments
4. Provide actionable recommendations through a dashboard.

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Things to take note of:

- Use Diabetes_and_Lifestyle_Dataset_ for this project.
- Domain understanding is very important. Before building your models research on the different diabetes types in the dataset (explore the feature diabetes_stage).

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Project Framework

You must follow the CRISP-DM Framework.

Modeling Instructions

Risk Classification (Primary Task)

- Predict patient risk class (Target – diabetes stage).
- Models – Decision Tree, Random Forest, XGBoost

Patient Segmentation

- Group patients based on lifestyle and health characteristics.
- Models - use k-means (with k=3).

Key Driver Analysis

- Identify what factors influence diabetes risk
- Methods – Use SHAP Values (Shapley Additive Explanations) for both risk classification and patient segmentation.

Additional Instructions

Version Control (Git & GitHub Requirements)

Each group must use GitHub for task management.

- Show meaningful commit history from all team members.

Your Repository must follow the Project Files Structure on BC Connect (note that this is a general guideline, adjust accordingly based on project)

Deliverables:

1. Technical Report following the CRISP-DM Framework as sections (maximum 2 pages, to be submitted on BC Connect, include the GitHub link and the Web Application Link). **NB: Clicking on the links should enable one to be directed to the relevant places.**
2. GitHub Repository which is properly structured with meaningful commit history and adequate documentation.
3. Interactive DASH Web App. Deploy your web application and ML model(s) as a web service using a platform like Render.
4. **Video submission**
   You need to submit a video or Onedrive link in a PDF document (**should you choose to submit a Onedrive link include it in the Technical Report document**) to a video that will show how you have met all the project criteria. Marks will **only** be awarded for the video as we are your client and will not be marking your code. Even if you make a video just about your code, you will not get a lot of marks because we want to see a working project after you explain your code. No time limit on the video but make sure you show all the criteria as fast as possible. Note that the video should cover:
   - Problem understanding
   - Data insights
   - Model results
   - Demo of the web app
   - Recommendations and conclusion.

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